

2.To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction

Code:-

```
#include

<stdio.h>

#include

<string.h>

#include

<stdlib.h>

int main()

{

    char input[MAX_INPUT];

    printf("=====\n");

    printf("    Welcome to Mini

Shell\n"); printf("Type 'exit' to

quit.\n");

    printf("=====\n");

    while (1)

    {

        printf("MiniShell> ");

        fflush(stdout);

        if (fgets(input, sizeof(input), stdin) == NULL)

        {
```

```
        printf("\nExiting
        shell...\n"); break;
    }
    input[strcspn(input, "\n")] =
    '\0'; if (strlen(input) == 0)
    {
        continue;
    }
    if (strcmp(input, "exit") == 0)
    {
        printf("Goodbye!\n
        n"); break;
    }
    printf("Command entered: %s\n", input);
}
Return 0;
}
OUTPUT:-
```

```
skill — nano skillq2.c — 80x24
UW PICO 5.09 File: skillq2.c Modified

    {
        history[count] = (char *)malloc(strlen(input) + 1);
        strcpy(history[count], input);
        count++;
    }

    printf("Executed: %s\n", input);
}

for (int i = 0; i < count; i++)
    free(history[i]);

free(history);
free(input);

printf("Memory released successfully.\n");

return 0;
}

^G Get Help ^O WriteOut ^R Read File ^Y Prev Pg ^K Cut Text ^C Cur Pos
^X Exit ^J Justify ^W Where is ^V Next Pg ^U UnCut Text ^T To Spell

apple@apples-MacBook-Pro-2 skill % ./skillq2
===== Mini Shell =====
Type 'history' to view history.
Type '!!!' to execute last command.
Type '!n' to execute command number n.
Type 'exit' to quit.

MiniShell> histtory
Executed: histtory
MiniShell> history

Command History:
1 : histtory

MiniShell> pwd
Executed: pwd
MiniShell> !!
Previous Command: pwd
MiniShell>
```